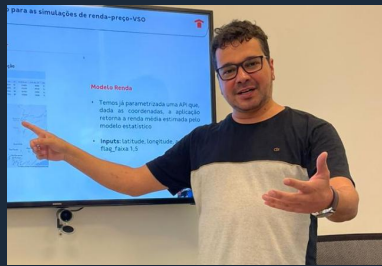


LUCAS A. SOUZA

Data Scientist | PhD in Physics



Data Scientist with a PhD in Physics and a strong foundation in applied mathematics. I transform complex datasets into actionable business insights through statistical modeling, machine learning, and strategic analysis. Experienced in bridging technical rigor with business goals to drive data-informed decisions.

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Technical Expertise

- Quantitative Modeling & Statistical Inference
- Machine Learning & Time Series Analysis
- Probabilistic Modeling & Simulation
- Scientific Computing & Numerical Methods
- Pricing Analytics, Market Intelligence & Monitoring Dashboards
- Causal & Temporal Analysis

Technical Stack

- Python (pandas, NumPy, scikit-learn, LightGBM, PyTorch)
- SQL & Analytical Data Processing
- Jupyter, Streamlit & Scientific Python Ecosystem
- Linux & Reproducible Research Workflows
- AI Agents, LLM Workflows & Agentic Automation

Research Interests

- Quantitative Finance & Algorithmic Trading
- Scientific Machine Learning

Education

2017 **Doctorate in Physics**

ITA , Brazil

2011 **Master's degree in Physics**

UFSJ, Brazil

2008 **Physics Undergraduate**

UFSJ, Brazil

Data Science Experience

9/2021 - **Data Scientist – Specialist level**

Tenda, Brazil

present

- Leads quantitative modeling initiatives for pricing, demand forecasting, customer behavior, and market intelligence across large-scale residential real estate operations.
- Designs predictive and probabilistic models to support pricing sophistication, including customer income distribution, financing behavior, subsidy allocation dynamics, and affordability analysis.
- Builds statistical and machine learning pipelines integrating geospatial, socioeconomic, commercial, and transactional data to support project-level pricing strategy and market analysis.
- Develops simulation frameworks, analytical tools, and monitoring systems for pricing execution, commercial performance, customer affordability dynamics, and identification of revenue capture opportunities.

Research Experience

2020 **Postdoc researcher**

ITA , Brazil

- Investigation on short-range correlations in nuclear matter of neutron stars.

2019 **Postdoc researcher**

USP, Brazil

- Theoretical study on nuclear reactions of neutron-rich nuclei.

Selected Professional Training (notes and scripts → [Github](#))

2025 **Statistics for Data Science and Machine Learning**

(ptBR) 20h

IA Expert Academy, Udemyl

2022 **TensorFlow: Machine Learning and Deep Learning with Python**

(ptBR) 19h

IA Expert Academy

2022 **Genetic Algorithm in Python** (ptBR) 4,5h

IA Expert Academy

2021 **The Complete SQL Bootcamp 2022** (En) 9h

Udemyl

2021 **Machine Learning and Data Science with Python from A to Z** (ptBR) 42h

IA Expert Academy, Udemyl

2017 **School on few-body physics** (En) 60h

MPIPKS, Dresden, Germany 

Teaching Experience (2/5 → [CV Lattes](#))

2019 **Professor of Physics and Math, 30h**

Unitau, Brazil

2018 **Temporary Professor of Physics, 40h**

UFLA, Brazil

Relevant Publications (2/14 → [Google Scholar](#))

- Pilling, Pazzianotto, Lucas A. Souza, [The Astrophysical Journal](#), **921**, 116(2021)
- Lucas A. Souza, Chimanski and Carlson, [Phys. Rev. C](#), **104**, 034623(2021)